

Mediator-Based Reward Design in Online Contextual Bandit

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Abstract

In Reinforcement Learning (RL), different reward functions may lead to the same optimal policy, while some reward functions can be substantially easier to learn. This paper proposes a framework that constructs surrogate rewards based on mediators between actions and rewards, informed by expert-provided causal directed acyclic graphs (DAGs). These DAGs encode domain knowledge from scientists. We show that our surrogate reward is unbiased and has reduced variance compared to the original reward when the mediator fully captures all causal pathways from the action to the reward. We further introduce an online reward-design agent that adaptively learns a surrogate reward in an unknown environment. We show that this reward-design agent can improve the regret guarantees of an online contextual bandit algorithm. Furthermore, our framework highlights improvement even without the surrogacy assumption, when total step T is small compared to the error violating the surrogacy assumption. We complement the theoretical analysis with simulation studies, demonstrating performance improvement even when the mediator only captures 56% of the causal effect.

1 Introduction

Reinforcement learning (RL), in which an agent sequentially interacts with an unknown environment and learns to maximize cumulative rewards, is widely used for solving decision-making problems. The reward function, often a mapping from the environment’s state to a scalar, is a crucial part of RL, as it determines the agent’s learning signal. Although many RL problems have a pre-specified reward function, prior literature (Laud 2004; Eschmann 2021) has shown that a well-designed reward function can lead to faster learning while maintaining the same optimal policy.

This paper addresses a common challenge in reward design: how can we design a reward function to reduce the variance in learning when the original reward to be maximized is **noisy**? This problem is motivated by real-world applications like mobile health (Trella et al. 2023; Ghosh et al. 2024) and advertising (Li et al. 2010; Schwartz, Bradlow, and Fader 2017), where the reward is defined by highly noisy human behaviors. Figure 1 illustrates an example in smoking cessation (Battalio et al. 2021), where a mobile app delivers digital interventions to prompt users’ stress management behavior. These interventions reduce the stress level, thereby decreasing the odds of subsequent smoking behavior. In this

scenario, the causal effect between prompts and stress levels is often less noisy than that between stress levels and smoking behavior.

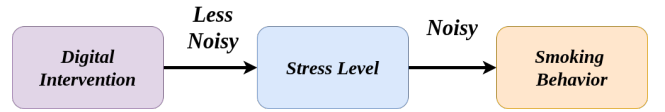


Figure 1: A noisy reward example in mobile health, where a digital intervention reduces smoking behavior by prompting stress management behavior, whose effect is mediated by stress level.

In domains like behavior psychology, the domain knowledge is often available in the form of causal directed acyclic graphs (DAGs) representing causal relations between variables. These causal relations are usually the domain experts’ knowledge of the major causal effects between variables in the system (Gopnik et al. 2004), and they may be included in the design of RL algorithms to improve the learning efficiency (Deng et al. 2023; Bareinboim, Zhang, and Lee 2024), especially in scientific fields (Yang et al. 2024; Gao et al. 2025). This paper aims to design surrogate rewards based on the **mediators**, the variables that mediate causal paths between actions and rewards. For example, the *stress level* is the mediator between digital interventions and smoking behavior in Figure 1.

Contributions. Our contributions are as follows

1. We propose a framework for mediator-based reward design in contextual bandits. Under the surrogacy assumption—that the reward is independent of the action given the mediator and state—we prove that our surrogate reward is unbiased with respect to the true expected reward and has a strictly lower variance.
2. Further, we propose an online reward design agent that adaptively learns the target mediator-based reward function and integrates it with any existing online bandit oracle for online decision-making.
3. We show theoretically that, when employing an adversarial bandit oracle to manage reward non-stationarity, our method achieves tighter regret bounds compared to standard linear UCB algorithms, with or without the surrogacy assumption.

4. We validate the performance improvement with simulation experiments.

2 Related Work

Reward Design. The term **reward engineering** is used interchangeably with **reward design** sometimes (Hirtz et al. 2024), where both terms refer to the creation and modification of reward functions to align the learned policy with the goal of the task. However, in other literature (Gupta et al. 2022; Ibrahim et al. 2024), **reward engineering** refers to creating a reward function that aligns with the human intent. Our method differs from the latter setting as we directly observe rewards. **Reward shaping** involves modifying the reward function to guide exploration or reduce variance in the learning process, without changing the optimal policy (Laud 2004). Existing methods such as potential-based reward shaping (Ng, Harada, and Russell 1999) and dense reward functions that allocate a sparse reward into multiple steps (Hare 2019; Eschmann 2021) guide exploration but rarely address high-noise scenarios, which is the central issue tackled in this paper. Our work uniquely focuses on mediator-based surrogate rewards to reduce variance from immediate noisy rewards.

Reward Design for Noisy Data There are works (Natarajan et al. 2013; Scott, Blanchard, and Handy 2013; Wang, Liu, and Li 2020) focusing on the definition of unbiased surrogate rewards, where the knowledge of the **noise** are used to recover the true loss. But these works are mainly concerned about eliminating bias instead of **variance reduction**. What’s more, the structural knowledge in the generation process of the reward has not garnered adequate attention.

Reward Design with Structural Knowledge. Mohan, Zhang, and Lindauer provide a comprehensive survey on accelerating the learning of RL through structural knowledge. Our proposed method broadly falls into the range of incorporating side information. Icarte et al. proposed a reward shaping utilizing structural information but provided no theoretical explanation and posed requirements about the reward function specification. On the other hand, we propose to exploit a common format for depicting structural information in causal inference, Directed Acyclic Graphs.

Causal Directed Acyclic Graphs and Mediator. Causal Directed Acyclic Graphs (DAGs) are constructed to depict prior knowledge about specific causal systems (Digitale, Martin, and Glymour 2022). DAGs have been used in causal problems to inform study design, statistical analysis, and algorithm improvement (Luis 2020; Thost and Chen 2021). Mediator, as defined by Baron and Kenny, is a variable that captures most part of relation from an independent variable to its dependent variable. A more straightforward definition is given in Figure 2.

The mediator-based surrogate index was introduced by Athey et al. (2019) to estimate the average treatment effect under the surrogacy assumption, with the primary outcome unobserved in the experimental sample and the treatment variable unobserved in the observational sample. In policy

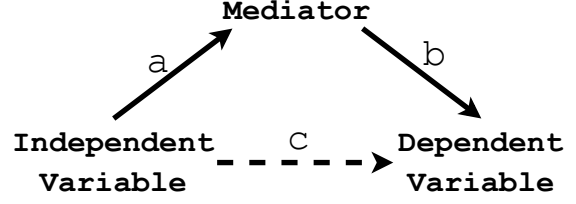


Figure 2: A variable functions as a mediator when it meets the following conditions: (a) variations in levels of the independent variable significantly account for variations in the presumed mediator (i.e., Path a), (b) variations in the mediator significantly account for variations in the dependent variable (i.e., Path b), and (c) when Paths a and b are controlled, a previously significant relation between the independent and dependent variables is no longer significant, with the strongest demonstration of mediation occurring when Path c is zero. (Baron and Kenny 1986)

learning, Yang et al. (2024) imputed missing long-term outcomes using the mediator-based surrogate index and maximized the imputed outcomes via off-policy optimization in a batched bandit setting. Our work differs from the above by adaptively learning the surrogate rewards online. **A key feature of our work is its modularity**—we develop an online reward design agent that can interface with any existing online bandit oracle, which is an agent making decisions in bandit environments; thus separating the reward design from the decision-making process. A technical challenge of this general framework is the non-stationarity in the surrogate reward due to the reward-learning process. We propose to use an adversarial bandit oracle to address this challenge.

3 Problem Setup

Notation. We use I_d and 0_d to denote the identity matrix and the null matrix of dimension $d \times d$ respectively. We use $\mathbf{0}$ to denote the null vector. We use $\|\cdot\|$ to denote the L_2 -norm. For a symmetric matrix A , we use $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$ to denote the smallest and the greatest eigenvalue of A respectively, and $\|\mathbf{x}\|_A$ to denote $\sqrt{\mathbf{x}^\top A \mathbf{x}}$ for a vector $\mathbf{x}$. A random vector $\mathbf{x} \sim SG_d(\sigma)$ is a d -dimensional sub-Gaussian variable such that, for any $\mathbf{u} \in \mathbb{R}^d$ with $\|\mathbf{u}\| = 1$, $\mathbb{E}[\exp(\mathbf{u}^\top \mathbf{x}t)] \leq \exp(\sigma^2 t^2 / 2), \forall t \in \mathbb{R}$.

Contextual bandit with mediators. We consider a contextual bandit problem where the agent at each step $t = 1, \dots, T$ observes context $\mathbf{S}_t \in \mathcal{S} \subset \mathbb{R}^{d_S}$, based on which it chooses an action $A_t \in \mathcal{A}$. The environment reveals the mediator $\mathbf{M}_t \in \mathbb{R}^{d_M}$ and the reward $R_t \in \mathbb{R}$ jointly from a distribution $P(\cdot | \mathbf{S}_t, A_t)$, where each $P(\cdot | \mathbf{s}, a)$ is a probability distribution over $\mathbb{R}^{d_M+1}$. The agent aims to optimize the cumulative reward $\sum_{t=1}^T R_t$. We define the mean reward function $R(\mathbf{s}, a) = \mathbb{E}[R_t | A_t = a, \mathbf{S}_t = \mathbf{s}]$.

Mediator-based surrogate reward. Figure 3 demonstrates a typical DAG for contextual bandit problems. When the dashed line is absent, $\mathbf{M}_t$ blocks all causal paths from

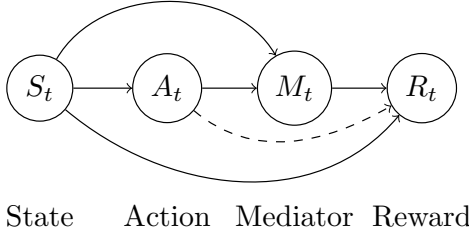


Figure 3: A DAG for contextual linear bandit with mediators. The surrogacy assumption is violated **when the dashed line between R and A exists**.

A_t to R_t . In the causal inference literature, this is called surrogacy (Assumption 1). In our theoretical and numerical analysis, we will **allow weak violation of the surrogacy assumption** to accommodate different real world scenarios. Although it is implicitly assumed in general, we emphasize that **causal sufficiency** is assumed in our DAG, which means that for a set of variables V , every common cause of any pair of variables in V is also in V (Spirtes, Glymour, and Scheines 2001).

Assumption 1 (Surrogacy, (Prentice 1989)). *The reward is independent of the action given the mediator and the state, i.e. $R_t \perp A_t \mid M_t, S_t$.*

We aim to design a surrogate reward $\tilde{R}_t = f(M_t, S_t)$ for some function $f : \mathbb{R}^{d_M} \times \mathcal{S} \mapsto \mathbb{R}$. We show in Proposition 1 that under surrogacy, $\tilde{R}_t = \mathbb{E}[R_t \mid M_t, S_t]$ is an unbiased surrogate reward having lower variance than the true reward.

Proposition 1 (Unbiasedness under surrogacy). *Define $\tilde{R}^*(m, s) = \mathbb{E}[R_t \mid M_t = m, S_t = s]$, the mean function of $\tilde{R}_t$ given $M_t = m$ and $S_t = s$. If Assumption 1 holds, we have*

$$\begin{aligned} & \mathbb{E}[R_t \mid A_t = a, S_t = s] \\ &= \mathbb{E}[\tilde{R}^*(M_t, S_t) \mid A_t = a, S_t = s], \end{aligned} \quad (1)$$

$$\begin{aligned} & \text{Var}(R_t \mid A_t = a, S_t = s) \\ & \geq \text{Var}(\tilde{R}^*(M_t, S_t) \mid A_t = a, S_t = s) \end{aligned} \quad (2)$$

for any $a \in \mathcal{A}$ and $s \in \mathcal{S}$. The inequality strictly holds if and only if there exists some subset $\mathcal{M} \subseteq \text{supp}(M_t \mid A_t = a, S_t = s)$ with $P(\mathcal{M}) > 0$, such that $\text{Var}(R_t \mid M_t = m, A_t = a, S_t = s) > 0$ for all $m \in \mathcal{M}$.

Proof. For any $a \in \mathcal{A}$ and $s \in \mathcal{S}$,

$$\begin{aligned} & \mathbb{E}[\tilde{R}^*(M_t, s) \mid A_t = a, S_t = s] \\ &= \mathbb{E}[\mathbb{E}[R_t \mid M_t, S_t = s] \mid A_t = a, S_t = s] \\ &= \mathbb{E}[\mathbb{E}[R_t \mid M_t, S_t = s, A_t = a] \mid A_t = a, S_t = s] \\ & \quad \quad \quad \text{(Assumption 1)} \\ &= \mathbb{E}[R_t \mid S_t = s, A_t = a]. \quad \text{(Law of total expectation)} \end{aligned}$$

The lower variance is straightforward by noting that

$$\begin{aligned} & \text{Var}(R_t \mid A_t = a, S_t = s) \\ &= \text{Var}(\mathbb{E}[R_t \mid M_t, A_t = a, S_t = s] \mid A_t = a, S_t = s) \\ & \quad + \mathbb{E}[\text{Var}(R_t \mid M_t, A_t = a, S_t = s) \mid A_t = a, S_t = s] \\ & \quad \quad \quad \text{(Law of total variance)} \\ & \geq \text{Var}(\mathbb{E}[R_t \mid M_t, A_t = a, S_t = s] \mid A_t = a, S_t = s) \\ &= \text{Var}(\tilde{R}^*(M_t, s) \mid A_t = a, S_t = s). \end{aligned}$$

If there exists $m \in \text{supp}(M_t \mid A_t = a, S_t = s)$, such that $\text{Var}(R_t \mid M_t = m, A_t = a, S_t = s) > 0$, then we have $\mathbb{E}[\text{Var}(R_t \mid M_t, A_t = a, S_t = s) \mid A_t = a, S_t = s] > 0$ and thus the inequality strictly holds. $\square$

A linear working model. Throughout the paper, we make an additional linear assumption about the reward and mediator generating processes:

$$R_t = \theta_{A_t}^\top S_t + \theta_S^\top S_t + \theta_M^\top M_t + \epsilon_t, \quad (3)$$

$$M_t = \Gamma_{A_t} S_t + \omega_t, \quad (4)$$

where $\theta_a \in \mathbb{R}^{d_S}$, $\Gamma_a \in \mathbb{R}^{d_M \times d_S}$ for each $a \in \mathcal{A}$, $\theta_S \in \mathbb{R}^{d_S}$, $\theta_M \in \mathbb{R}^{d_M}$, and $\epsilon_t \sim SG_1(\sigma_\epsilon)$, $\omega_t \sim SG_{d_M}(\sigma_\omega)$. Note that we have θ_{A_t} here to **allow violation of the surrogacy assumption**, which holds when $\theta_a \equiv 0$ for all $a \in \mathcal{A}$.

Policy and regret. We define a policy π as a mapping from context to action, i.e., $\pi : \mathcal{S} \mapsto \mathcal{A}$. The performance of an online algorithm is measured by regret, which is defined as the difference between the expected cumulative rewards generated by the optimal policy and the expected cumulative rewards generated by the algorithm. Assuming $\pi^*(s) := \arg \max_a R(s, a)$ is the optimal policy, we define the regret as

$$\text{Regret}_T := \sum_{t=1}^T (R(S_t, \pi^*(S_t)) - R(S_t, A_t)). \quad (5)$$

4 Mediator-based Online Reward Learning

Our proposed solution has two critical components, an **online reward design agent** that adaptively learns a reward mapping based on S_t and M_t , and an online bandit oracle. The interaction between the online reward design agent and the online bandit oracle is characterized in Figure 4.

4.1 Online Reward Design Agent

The **online reward design agent** is learning the function $\tilde{R}^*(m, s) = \mathbb{E}[R_t \mid M_t = m, S_t = s] = s^\top \theta_S + m^\top \theta_M$, where θ_S and θ_M are the coefficients of the linear model in (3).

We use online ridge regression to learn the coefficients of the linear model. At each decision time t , the estimators are

$$\begin{bmatrix} \hat{\theta}_{S,t} \\ \hat{\theta}_{M,t} \end{bmatrix} = \left(\sum_{\tau=1}^{t-1} X_\tau X_\tau^\top + \lambda I_{d_S+d_M} \right)^{-1} \sum_{\tau=1}^{t-1} X_\tau R_\tau, \quad (6)$$

where λ is a tuning parameter and $X_t = (S_t^\top, M_t^\top)^\top$ denotes the regression covariates. The overall process is described in Algorithm 1.

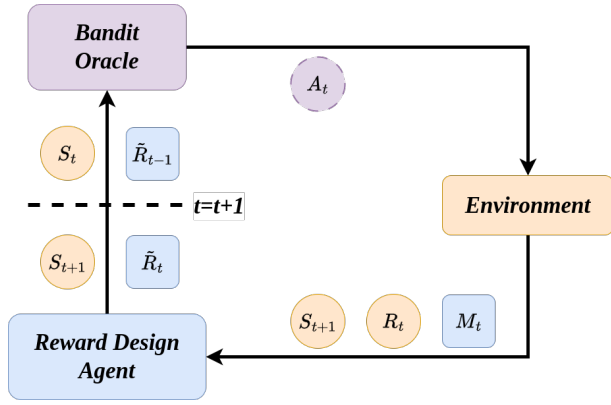


Figure 4: At each decision time t , the oracle observes the state S_t , chooses an action $A_t \in \mathcal{A}$, and receives the surrogate reward $\tilde{R}_t$. Variables in **orange (circle)** are observed variables in the standard bandit setting, and those in **blue (rounded square)** are the additional variables introduced in this framework for surrogate reward design. Action is in **purple (dashed circle)** to show that it's dependent on the oracle.

Algorithm 1: Mediator-based Online Reward Learning

Input: online bandit oracle $\mathcal{O}$; dataset for reward design $\mathcal{D}_0 = \{\}$; ridge regularization parameter λ .

- 1: Initialize π_1 by the uniform random policy.
- 2: **for** $t = 1, \dots, T$ **do**
- 3: Observe current state S_t
- 4: Sample $A_t \sim \pi_t(S_t)$, and environment generates $(R_t, M_t) \sim P(\cdot | S_t, A_t)$
- 5: Add observation for reward design $\mathcal{D}_t = \mathcal{D}_{t-1} \cup \{(S_t, M_t, R_t)\}$
- 6: Run ridge regression in (6) on $\mathcal{D}_t$ to get estimators $\hat{\theta}_{S,t}$ and $\hat{\theta}_{M,t}$
- 7: Construct reward $\tilde{R}_t = \hat{\theta}_{S,t}^\top S_t + \hat{\theta}_{M,t}^\top M_t$
- 8: Query bandit oracle $\pi_{t+1} = \mathcal{O}(S_t, A_t, \tilde{R}_t)$
- 9: **end for**

4.2 The Need for an Adversarial Bandit Oracle

The **online bandit oracle** is a bandit algorithm that at each decision time t takes a tuple $(S_t, A_t, \tilde{R}_t)$ as the input and outputs the next policy π_{t+1} , a mapping from the state space $\mathcal{S}$ to a distribution over the action space $\mathcal{A}$.

However, the standard online contextual bandit problem assumes that the reward function is stationary, i.e. $\mathbb{E}[R_t | S_t = s, A_t = a, \mathcal{H}_t] \equiv R(s, a)$ for all t and some fixed function $R : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$. Here $\mathcal{H}_t$ is the whole history up to time t . This stationarity is violated due to the learning process of the *online reward design agent*. Indeed, the oracle learns on the signal $\tilde{R}_t$ and $\mathbb{E}[\tilde{R}_t | S_t = s, A_t = a, \mathcal{H}_t] = \hat{\theta}_{S,t}^\top s + \hat{\theta}_{M,t}^\top \mathbb{E}[M_t | S_t = s, A_t = a]$ is non-stationary due to the changing $\hat{\theta}_{S,t}$ and $\hat{\theta}_{M,t}$. To overcome this issue, we propose to use adversarial bandit oracles (Auer et al. 2002; Chu et al. 2011) that adapt to the changing reward signal.

Properties of the Adversarial Bandit Oracle. We say that an online bandit oracle $\mathcal{O}$ is an adversarial oracle if it satisfies the following regret guarantee in Assumption 2. In contrast to the standard bandit setting, the adversarial bandit oracle targets the adversarial regret guarantee that is defined as the cumulative gaps between the optimal fixed policy and the policy chosen by the learner:

$$\text{AdvRegret}_T :=$$

$$\max_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{A \sim \pi(\cdot | S_t)} [\tilde{R}_t(S_t, A)] - \tilde{R}_t(S_t, A_t) \right],$$

where $\tilde{R}_t(S_t, a) = \mathbb{E}[\tilde{R}_t | S_t, A_t = a]$ is the mean reward at step t . In the adversarial bandit setting, it is often the case that the mean reward is chosen by an adversary, which may depend arbitrarily on the history.

Since the average of surrogate reward function $\tilde{R}_t(s, a)$ converges to the true surrogate reward function $\tilde{R}^*(s, a) := \theta_S^\top s + \theta_M^\top \mathbb{E}[M_t | S_t = s, A_t = a]$ as the number of steps goes to infinity, the adversarial regret guarantee converges to the standard bandit regret guarantee w.r.t the true reward function R^* . Thus, our goal is to use the adversarial bandit oracle with sublinear and variance-adaptive adversarial regret guarantee as in the following assumption.

Assumption 2 (Variance-adaptive adversarial bandit regret guarantee). *We assume that the adversarial bandit oracle has upper bound on the following regret, with a high probability $1 - \delta$,*

$$\text{AdvRegret}_T = \tilde{\mathcal{O}} \left(\sigma_R^2 \cdot \text{poly}(d_S, |\mathcal{A}|, \log(T/\delta)) \cdot T^{1-\alpha} \right),$$

where $\alpha \in [0, \frac{1}{2}]$, *poly* stands for polynomial functions, $\tilde{\mathcal{O}}$ hides the logarithmic terms, and $\sigma_R^2 := \text{Var}[R_t | S_t = s, A_t = a]$ is the variance of the reward signal.

To demonstrate the practicality of the assumption about the oracle, we note that there exist adversarial bandit algorithms (Rakhlin and Sridharan 2016; Neu and Olkhovskaya 2020; He et al. 2022) that already consider the case where the mean reward is arbitrarily chosen by an adversary. Algorithms that assume a strict linear reward can be a candidate for our oracle (Syrkanis et al. 2016; Kuroki et al. 2024; Banihashem et al. 2024). These algorithms are developed for the case of a deterministic reward without variance adaptivity. In Appendix A, we discuss a reduction method proposed by Lattimore and Szepesvári, which absorbs the noise term as an additional dimension of the adversarial reward vector, to fit these algorithms into our framework.

4.3 Regret Analysis

We first introduce an additional regularity assumption concerning the boundedness of the coefficients in (3).

Assumption 3 (Bounded coefficients). *We assume that there exist constants $\mathcal{E}, C > 0$ s.t.*

$$\forall t \in \{1, 2, \dots, T\}, \|\mathbf{S}_t\| \leq 1 \text{ almost surely}, \quad (7)$$

$$\max_{a \in \mathcal{A}} \|\mathbf{T}_a\| \leq C, \max_{a \in \mathcal{A}} \|\boldsymbol{\theta}_a\| \leq \mathcal{E}C, \quad (8)$$

$$\|\boldsymbol{\theta}_M\| \leq C. \quad (9)$$

Note that **surrogate error** $\mathcal{E}$ controls the ratio of the effect of A_t on R_t not captured by $\mathbf{M}_t$ to that captured by $\mathbf{M}_t$. A greater $\mathcal{E}$ implies more severe violation of the surrogacy assumption. $\mathcal{E} = 0$ implies surrogacy (Assumption 1).

We now present the high-probability regret bound for Algorithm 1 when the adversarial online bandit oracle is chosen as *RealLinExp3* (Neu and Olkhovskaya 2020). The standard proof is extended to incorporate the noise of the reward.

Theorem 1 (Regret bound for Algorithm 1). *When Assumption 3 holds, the regret of Algorithm 1 is bounded with a high probability by*

$$\tilde{O}\left(\sigma_\omega \sqrt{d_S |\mathcal{A}| T} + \sigma_\epsilon \sqrt{(d_S + d_M) T} + \mathcal{E} T\right). \quad (10)$$

Now we compare our regret bound with the original bound (omitting the $\tilde{O}$ notation). The minimax regret bound for stochastic contextual linear bandits is $\sigma_\omega \sqrt{d_S |\mathcal{A}| T} + \sigma_\epsilon \sqrt{d_S |\mathcal{A}| T}$. To interpret the improvement, we note that our regret bound in (10) has a reduction in a multiplicative factor of $\sqrt{|\mathcal{A}|}$ at the price of an additional $\sqrt{d_M}$ term. This improvement is due to the fact that the ridge regression estimator pools information across all arms to learn the effect of the mediator on the reward, while naive contextual linear bandits with a discrete action set estimates the coefficient of each arm separately. Making this improvement requires an additional regression based on the mediator, thus leading to the additional $\sqrt{d_M}$ term. The linear term $\mathcal{E} T$ only appears if Assumption 1 is violated; this term comes from the bias in the surrogate reward due to the violation. Accordingly, even when there is a violation of Assumption 1, our method has a better regret bound when

$$\sqrt{T} \leq \frac{\sigma_\epsilon}{\mathcal{E}} (\sqrt{d_S |\mathcal{A}|} - \sqrt{d_S + d_M}). \quad (11)$$

The improvement is significant when:

1. $\sigma_\omega \ll \sigma_\epsilon$: the mediator generating process is much less noisy than the primary reward generating process;
2. $d_S \gg d_M$: the context is of a much higher dimension than the mediator;
3. large $|\mathcal{A}|$: we have a relatively large action set;
4. small $\mathcal{E}$: the action has a small uncaptured direct effect size on the reward.

5 Proof of Theorem 1

Proof Sketches. The regret in our framework can be attributed to two components: (1) The surrogate reward error incurred by the reward design agent, which accounts for both the cost of collecting structural information and the inaccuracy of the mediators. We bound the information cost using properties of the ℓ^2 -regularized least-squares estimator (Abbasi-Yadkori, Pál, and Szepesvári 2011; Tropp et al. 2015), and we capture the mediator’s inaccuracy with a linear term $\mathcal{E} T$. (2) The standard regret arising from the exploration–exploitation trade-off in the bandit setting, for which extensive analyses exist in the online learning literature (Neu and Olkhovskaya 2020).

Due to the sub-Gaussianity of ϵ_t and ω_t , $\forall \delta \in (0, 1)$,

$$\begin{aligned} \mathbb{P}(|\epsilon_t| \geq z) &\leq 2 \exp\left(-\frac{z^2}{2\sigma_\epsilon^2}\right), \forall z \in \mathbb{R}^+ \\ \implies \mathbb{P}\left(\max_{t=1, \dots, T} |\epsilon_t| \leq \sigma_\epsilon \sqrt{2 \log(2T/\delta)}\right) &\geq 1 - \delta. \\ \mathbb{P}\left(\max_{\|\mathbf{u}\| \leq 1} |\mathbf{u}^\top \omega_t| \leq \sigma_\omega \sqrt{2 \log(2d_M/\delta)}\right) &\geq 1 - \delta. \end{aligned}$$

To simplify notation we denote

$$R_\omega := \sigma_\omega \sqrt{2 \log(2d_M/\delta)} = \tilde{O}(\sigma_\omega).$$

5.1 Concentration for Sample Covariance Matrix

We first introduce and prove a proposition to be used later. Let $d := d_S + d_M$.

Proposition 2 (Sample Covariance Matrix Concentration). *In a contextual linear bandit problem, suppose action $a \in \mathcal{A}$ is chosen $T_a \in \mathbb{N}$ times after step T . So we have get $|\mathcal{A}|$ groups of observations X . The group of a contains T_a observations, whose i -th observation is denoted as $X_i^{(a)}$. For any $\lambda > 0$,*

$$\lambda_{\min}(\lambda I + \sum_{t=1}^T X_t X_t^\top) = \tilde{\Theta}(T).$$

Proof. Let Σ_S and Σ_ω be the covariance matrix of states S and mediator noises ω . We denote

$$\begin{aligned} V_T &:= \lambda I + \sum_{t=1}^T X_t X_t^\top, \\ U_a &:= \begin{bmatrix} I_{d_S} \\ \mathbf{\Gamma}_a \end{bmatrix} \in \mathbb{R}^{d \times d_S}, \\ \Sigma_a &:= U_a \Sigma_S U_a^\top + \text{diag}\{0_{d_S}, \Sigma_\omega\}. \end{aligned}$$

Since $X^{(a)} \sim \mathcal{N}(\mathbf{0}, \Sigma_a)$ and assuming that policy converges to π , we can rewrite X as i.i.d variables as $X \sim \mathcal{N}(\mathbf{0}, \Sigma)$, where $\Sigma := \int_{s \in \mathcal{S}} P(s) \Sigma_{\pi(s)} ds$.

Assuming $\lambda_0 = \lambda_{\min}(\Sigma) > 0$, we know that

$$\begin{aligned} \sigma^2 &:= \lambda_{\max}(\Sigma) \\ &\leq \max \left[\max_{a \in \mathcal{A}} (\lambda_{\max}(U_a \Sigma_S U_a^\top)), \lambda_{\max}(\Sigma_\omega) \right], \end{aligned}$$

which implies $X \sim SG_d(\sigma)$.

We know that

$$\lambda_{\min}(V_T) \geq T \lambda_{\min} \left(\underbrace{\frac{\sum_{t=1}^T X_t X_t^\top}{T}}_{\text{Noted as } \hat{\Sigma}} \right).$$

According to Bernstein inequality on sample covariance matrix (Tropp et al. 2015), we have

$$|\lambda_{\min}(\hat{\Sigma}) - \lambda_{\min}(\Sigma)| \leq \rho(\hat{\Sigma} - \Sigma) = \tilde{\Theta} \left(\sigma^2 \sqrt{\frac{d}{T}} \right),$$

which implies

$$\lambda_{\min}(\hat{\Sigma}) \geq \lambda_0 - \tilde{\Theta} \left(\sigma^2 \sqrt{\frac{d}{T}} \right).$$

Since $\sigma^2 \sqrt{dT} = o(T)$ and $\lambda_{\min}(V_T) \geq T \lambda_{\min}(\hat{\Sigma})$ we have

$$\lambda_{\min}(V_T) \geq \tilde{\Theta}(T - \sigma^2 \sqrt{dT}) = \tilde{\Theta}(T).$$

□

5.2 Surrogate Reward Error

In this section, we bound the regret incurred by using the surrogate reward designed by the reward design agent. We begin by analyzing the general properties of the estimator given the true covariance matrix, and then apply Proposition 2 to estimate the sample covariance matrix in this environment.

Here, we draw your attention to the two notations that we use:

1.

$$\begin{aligned} R(\mathbf{s}, a) &:= \mathbb{E}[R_t \mid A_t = a, \mathbf{S}_t = \mathbf{s}] \\ &= (\boldsymbol{\theta}_S^\top + \boldsymbol{\theta}_M^\top \boldsymbol{\Gamma}_a + \boldsymbol{\theta}_a^\top) \mathbf{s} \end{aligned}$$

is the expectation of true reward for the whole problem, which cannot be known exactly because our observation is corrupted by noise.

2. $r_t(\mathbf{s}, a) := \hat{\boldsymbol{\theta}}_{M,t}^\top \boldsymbol{\Gamma}_a \mathbf{s} + \hat{\boldsymbol{\theta}}_{M,t}^\top \boldsymbol{\omega}_t$ is an estimation of the contribution from action a to $R(\mathbf{s}, a)$ at decision time t . This is the surrogate reward we feed to the oracle since it only determines the selection of action. And we have $|r_t(\mathbf{s}, a)| \leq C^2 + CR_\omega$ according to Assumption 3.

We first analyze the error incurred when using r_t to replace the true contribution of a in R . We define

$$\begin{aligned} e_t(\mathbf{s}, a) &= (\boldsymbol{\theta}_M^\top \boldsymbol{\Gamma}_a + \boldsymbol{\theta}_a^\top) \mathbf{s} - \mathbb{E}[r_t(\mathbf{s}, a)] \\ &= (\boldsymbol{\theta}_M - \hat{\boldsymbol{\theta}}_{M,t})^\top \boldsymbol{\Gamma}_a \mathbf{s} + \boldsymbol{\theta}_a^\top \mathbf{s} \\ &\leq \sup_{\mathbf{s}, a} \|\boldsymbol{\Gamma}_a \mathbf{s}\| \cdot \|\boldsymbol{\theta}_M - \hat{\boldsymbol{\theta}}_{M,t}\| + \|\boldsymbol{\theta}_a\| \|\mathbf{s}\| \\ &\leq C \|\boldsymbol{\theta}_M - \hat{\boldsymbol{\theta}}_{M,t}\| + \mathcal{E}C. \quad (\text{Assumption 3}) \end{aligned}$$

We assume that after a sufficient time t , the policy converges to π , and we denote

$$\begin{aligned} \bar{\boldsymbol{\theta}}_a &:= \mathbb{E}[\boldsymbol{\theta}_a \mid \pi], \\ X_t &:= \begin{bmatrix} \mathbf{S}_t \\ \mathbf{M}_t \end{bmatrix} \in \mathbb{R}^d. \end{aligned}$$

Then for $R_t = [(\boldsymbol{\theta}_S + \boldsymbol{\theta}_{a_t})^\top \mathbf{S}_t + \boldsymbol{\theta}_M^\top \mathbf{M}_t] + \epsilon_t$, there exists $\boldsymbol{\theta}^* := (\boldsymbol{\theta}_S + \bar{\boldsymbol{\theta}}_a, \boldsymbol{\theta}_M)$ such that $\epsilon_t = R_t - \boldsymbol{\theta}^{*\top} X_t \sim SG_1(\sigma_\epsilon)$ and the sequence $\{\epsilon_t\}$ is a martingale difference process. And $\|X_t\|$ is bounded because $\|\mathbf{S}_t\|$ is bounded. Let $\|X_t\| \leq L$.

Abbasi-Yadkori, Pál, and Szepesvári proved that, when use the l^2 -regularized least-squares parameter estimate $\hat{\boldsymbol{\theta}}_T$

with regularization coefficient $\lambda = 1/T$, with probability $1 - \delta$,

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|_{V_T} \leq \sigma_\epsilon \sqrt{d \log(1 + T^2 L) - d \log \delta} + \|\boldsymbol{\theta}^*\| / \sqrt{T},$$

Using Proposition 2, $\lambda_{\min}(V_T) \geq \tilde{\Theta}(T)$. We know that

$$\begin{aligned} &\|\boldsymbol{\theta}_M - \hat{\boldsymbol{\theta}}_{M,T}\| \\ &\leq \|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\| \\ &\leq \frac{1}{\sqrt{\lambda_{\min}(V_T)}} \|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\|_{V_T} \\ &= \tilde{\mathcal{O}}\left(\frac{\sigma_\epsilon \sqrt{d}}{\sqrt{T}}\right). \end{aligned}$$

And sum the error over total step T

$$\begin{aligned} &\mathbb{E} \left[\sum_{t=1}^T e_t(\mathbf{s}, a_t) \right] \\ &\leq C \cdot \mathbb{E} \left[\sum_{t=1}^T \|\hat{\boldsymbol{\theta}}_{M,t} - \boldsymbol{\theta}_M\| \right] + \mathcal{E}CT \\ &= C \cdot (\sigma_\epsilon \sqrt{d}) \tilde{\mathcal{O}}(\sqrt{T}) + \mathcal{E}CT. \end{aligned}$$

5.3 True Regret

Note that the Assumption 2 implies that $\pi^*(s) = \arg \max_{a \in \mathcal{A}} R(\mathbf{s}, a)$ also satisfies that

$$\begin{aligned} &\mathbb{E} \left[\sum_{t=1}^T (r_t(\mathbf{S}_t, \pi^*(\mathbf{S}_t)) - r_t(\mathbf{S}_t, A_t)) \right] \\ &= \tilde{\mathcal{O}}(\text{poly}(d_S, |\mathcal{A}|, \log(T/\delta)) \cdot T^{1-\alpha}). \end{aligned}$$

To proceed, we analyze the true regret w.r.t $r^*(a, s)$. We have

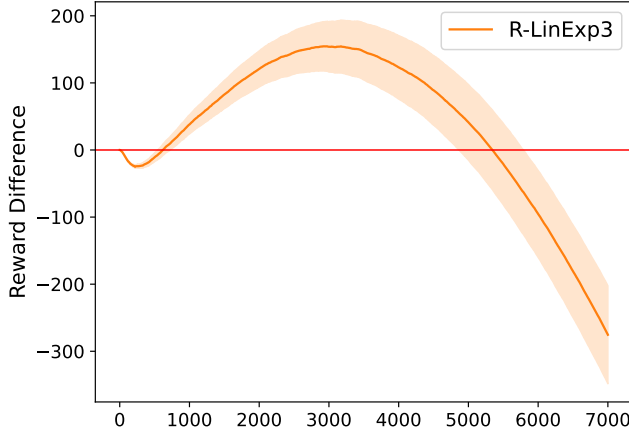
$$\begin{aligned} &\mathbb{E} \left[\sum_{t=1}^T (R(\mathbf{S}_t, \pi^*(\mathbf{S}_t)) - R(\mathbf{S}_t, A_t)) \right] \\ &\leq \mathbb{E} \left[\sum_{t=1}^T (r_t(\mathbf{S}_t, \pi^*(\mathbf{S}_t)) - r_t(\mathbf{S}_t, A_t)) \right] \\ &\quad + 2 \sup_{\mathbf{s}, a} \mathbb{E} \left[\sum_{t=1}^T e_t(\mathbf{s}, a) \right] \\ &= (C^2 + CR_\omega) \cdot \tilde{\mathcal{O}}(\text{poly}(d_S, |\mathcal{A}|, \log(T/\delta)) \cdot T^{1-\alpha}) \\ &\quad + C \cdot (\sigma_\epsilon \sqrt{d}) \tilde{\mathcal{O}}(\sqrt{T}) + \mathcal{E}CT. \end{aligned}$$

(Assumption 2 for the first term)

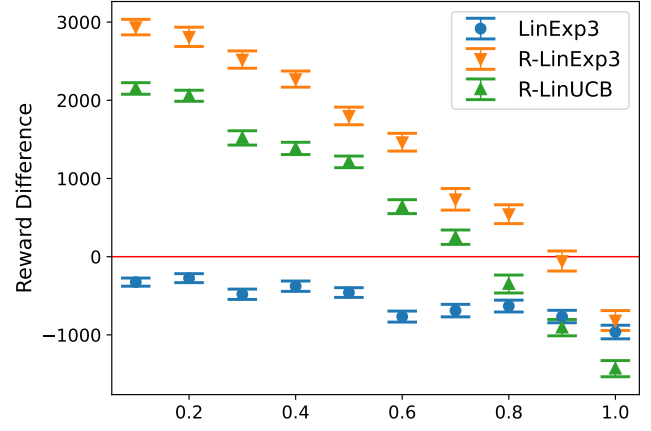
As established in Neu and Olkhovskaya (2020, Theorem 2), the polynomial term and exponent α are $\sqrt{d_S |\mathcal{A}| \log T}$ and $\frac{1}{2}$, respectively. Plugging these into the regret bound yields the final result.

6 Simulation Study

In this section, we conduct simulation studies on the contextual bandit with mediators to complement the theoretical results. The goal is to demonstrate the benefit of reward design agent, and the necessity of adversarial bandit oracle. The code is available in the supplementary material.



(a) Reward difference over time T . $\sigma_\omega = 0.1, \mathcal{E} = 2$.



(b) Reward difference vs. surrogate error $\mathcal{E}$. $\sigma_\omega = 0.01$.

Figure 5: Performance comparison of R-LinExp3, R-LinUCB, and LinExp3 relative to LinUCB. (a) plots cumulative reward difference as a function of time T ; (b) shows how increasing surrogate error $\mathcal{E}$ reduces the performance of reward-designed agents, consistent with Eq. (10). The y -axis in both panels indicates the reward difference relative to LinUCB. Error bars represent standard errors over 700 independent runs.

6.1 Setup

Environment. In the simulated bandit environment, we set the number of arms to $|\mathcal{A}| = 20$, with feature dimensions $d_S = 10$ and $d_M = 5$. The state vectors S_t are sampled uniformly from the unit ball. All coefficient vectors are drawn from normal distributions and rescaled to satisfy the norm constraint specified in Assumption 3, with norm bound $C = 2$. The surrogate error $\mathcal{E}$ varies by experimental setting. Noise terms are defined as $\omega_t \sim \mathcal{N}(\mathbf{0}, \sigma_\omega^2 I_{d_M})$ and $\epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2)$, where $\sigma_\epsilon = 10$ and σ_ω is case-specific. Each experimental run consists of 7,000 steps and is repeated 700 times.

Candidate Algorithms. The two base algorithms are LinExp3 (Neu and Olkhovskaya 2020), an adversarial bandit oracle, and LinUCB (Li et al. 2010), a stochastic bandit oracle. For each oracle, we test both how the naive version learns on the original reward, and the version with the reward design agent learns on the surrogate reward. We refer to the latter versions as R-LinExp3 and R-LinUCB, respectively.

Measurement. We characterize the performance of an algorithm by **reward difference** $R - R_0$, where R_0 is the expected reward collected by the baseline algorithm LinUCB and R is the expected reward collected by the algorithm to be analyzed. So a positive value denotes a performance improvement.

6.2 Results

In general, our method (R-LinExp3) achieves higher cumulative rewards than other candidate algorithms by effectively capturing additional structural information and demonstrating robustness to reasonable levels of surrogate error. Here, we focus on analyzing the effect of the **number of steps** T and the **surrogate error** $\mathcal{E}$.

Figure 5a depicts how the step T affects the performance. At the start, our algorithm’s performance is inferior to that of LinUCB because the reward design agent needs some data on which to “warm up”. Before warming up, the surrogate reward cannot accurately indicate the contribution of the chosen action. With a warm start, though, our algorithm outperforms LinUCB until T reaches a relatively high level, where the influence of bias preponderates over that of variance. This behavior is consistent with the condition in Eq. (11).

Figure 5b shows that an increase in surrogate error reduces the performance of the reward design agent, which is consistent with the outcome in Eq. (10). The fact that R-LinExp3 still achieves higher rewards even when $\mathcal{E}$ reaches 0.8—meaning the mediator captures only approximately $1/(1 + 0.8) \approx 56\%$ of the causal effect—demonstrates robustness to relatively high levels of surrogate error. Moreover, R-LinExp3 consistently outperforms R-LinUCB, supporting our theoretical motivation for adopting adversarial bandit oracles. Given that the underlying environment is stochastic, we would expect LinUCB to surpass LinExp3, which is reflected by the consistently negative values of the blue points.

7 Conclusion

In this paper, we propose a flexible framework for low-noise surrogate rewards design, exploiting the structural information provided by causal DAGs. We analyze the regret bound of our framework when an adversarial bandit oracle is used. From both theoretical and applicative views, we show significant improvements when high-quality (low-noise, high surrogacy) mediators are available. This framework accommodates real-world applications by harnessing the power of existing online-learning algorithms and domain knowledge.

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One can frame a stochastic linear environment as an adversarial linear one, by the following steps.

We define

$$\begin{aligned}\psi : \mathbb{R}^{d_S} &\longrightarrow \mathbb{R}^{d_S+1}, \\ s &\mapsto (s, 1),\end{aligned}$$

where the additional dimension allows us to incorporate the noise ϵ_t as a part of the parameter vector θ'_a of action a provided by the adversary, by letting $\theta'_{a_t} := (\theta_{a_t}, \epsilon_t)$. Thus $r_t := \langle S_t, \theta_{a_t} \rangle + \epsilon_t = \langle \psi(S_t), \theta'_{a_t} \rangle$. Note that action set $\mathcal{A}$ is not changed, while the adversarial environment chooses action vector θ_a in a higher-dimensional set.

A Reduction from stochastic linear bandits to adversarial linear bandits

Assume that in a stochastic linear environment, at each decision time t ,

1. The agent observes the state S_t and chooses an action $a_t \in \mathcal{A}$ accordingly.
2. The agent receives reward $r_t := \langle S_t, \theta_{a_t} \rangle + \epsilon_t$, where ϵ_t is a centered, bounded noise.